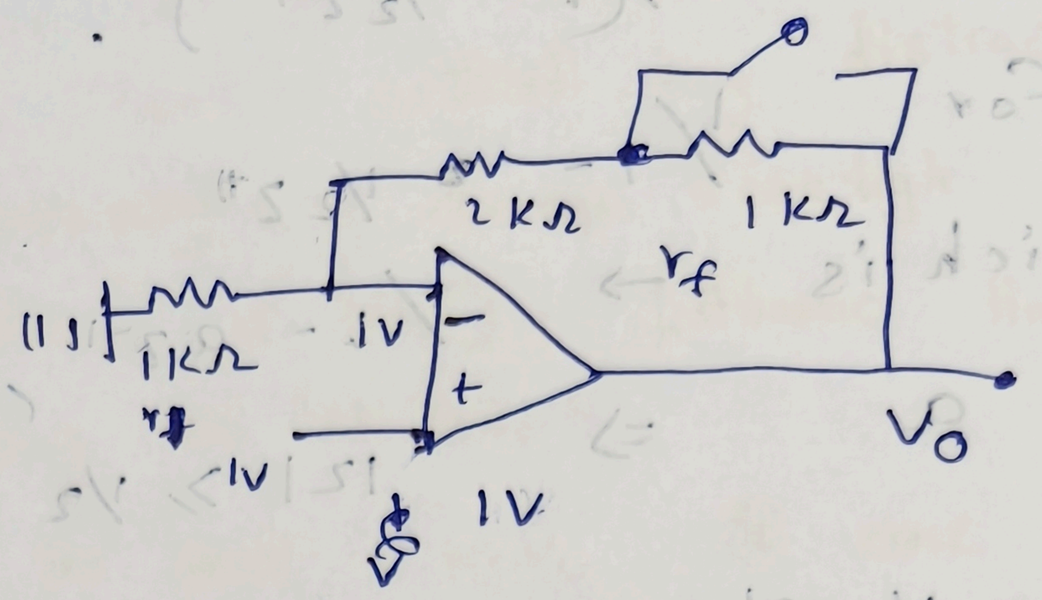


→ Given Ideal op-Amp.

⇒ $V_f = V_-$.

→ No current enters the V_f, V_- ∞ res ⇒ $I^+ = I^- = 0$.

⇒ Current through r_f

$r_f = 1+2 = 3k\Omega$

⇒ Drop on $r_1 = 1K\Omega \times 1 = 1V$

⇒ Drop on $r_f + r_1 = V_0$

↳ [1+2 = 3KΩ] ★

⇒ $\frac{V_0}{V_1} = \left(\frac{r_1}{r_f + r_1}\right)^{-1} = \frac{4}{1}$

→ Second case switch close, 1KΩ short ⇒ $r_f = 2K\Omega$.

⇒ $\frac{x}{V_1} = \left(\frac{r_1}{r_f + r_1}\right)^{-1} = \frac{3}{1}$

⇒ $\frac{x}{V_0} =$

$$\Rightarrow \frac{x}{V_0} = \frac{3}{4}$$

$$x = \frac{3 V_0}{4} .$$